ENTERPRISE INFORMATION ASSURANCE DASHBOARD

Pakistan Smart Railways (PSR) · CISO executive view · generated 2026-11-13

OVERALL IA STATUS

DEVELOPING

from governance maturity 1.9/5

Governance maturity	Overall risk score	High / Critical risks	Control effectiveness	Compliance	Audit findings
1.9 / 5	**11.9** / 25	**14** of 17	**12.0%**	**37.9%**	**13**
Weak	High	Act now	Critical	Weak	Act now
Year 3 target 3.7	inherent 13.8, target 7.2	5 Critical inherent	weighted 30.0%	22.75 of 60 requirement cells	3 Critical, 9 High

Critical control gaps	IAM readiness	BC/DR readiness	Incident readiness	AI governance readiness	AI automation impact
4	**40.0%**	**60.0%**	**45.5%**	**3.75** / 10	**81.4%**
Act now	Weak	Moderate	Weak	Weak	Strong
CTL-12, CTL-15, CTL-16, CTL-25	4 of 10 checks	6 of 10 checks	5 of 11 checks	4.5 of 12 checks	trust score 79.7%

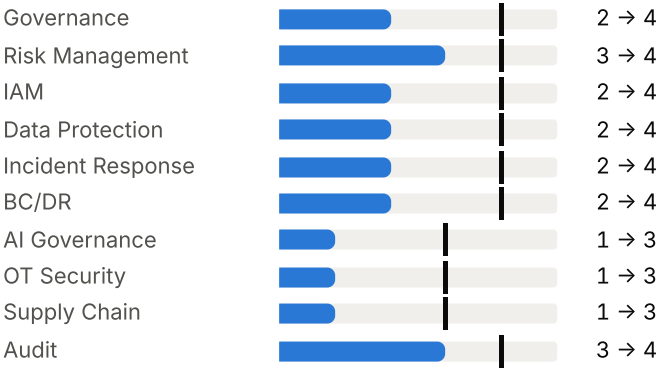
TOP 5 RISKS (CURRENT RESIDUAL)

1. **R-06** Broken access control in app and portal APIs · 16/25 Critical (inherent 16)
2. **R-03** Unpatched legacy and end-of-life OT · 15/25 High (inherent 20)
3. **R-05** Ransomware on ticketing and reservations · 15/25 High (inherent 20)
4. **R-09** Mass exfiltration of passenger data · 15/25 High (inherent 20)
5. **R-01** Compromise of the centralized train control system · 15/25 High (inherent 15)

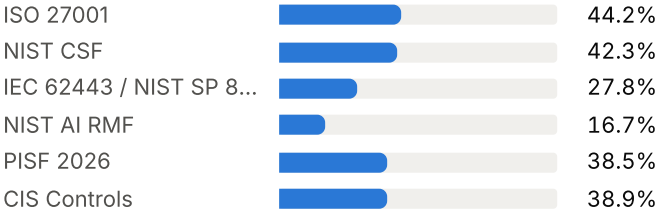
TOP 5 RECOMMENDATIONS

1. **CTL-07** Multi-factor authentication · Partial · owner CISO
2. **CTL-25** Incident and recovery exercise programme · Gap · owner CISO
3. **CTL-12** Application and API security testing · Gap · owner CIO
4. **CTL-02** Controlled OT remote access · Gap · owner CISO
5. **CTL-01** IT/OT network segmentation · Partial · owner OT/Engineering

GOVERNANCE MATURITY BY CAPABILITY (SCORE / 5, BAR = TARGET YEAR 3)



COMPLIANCE BY FRAMEWORK (A2 HEAT MAP, %)



AI AUTOMATION

Process	**Control mapping and gap analysis**
Method	**Ollama LLM (qwen2.5:7b) + retrieval**
Manual effort	**10.8 hrs**
AI assisted effort	**2.0 hrs**
Human approval	**Required**
AI Trust Score	**59/74 = 79.7%**
Accepted by reviewers	**59/74 = 79.7%**
Recall of manual mapping	**35.5%**
Hallucinated IDs blocked	**69**

Estimated improvement **81.4%**

Manual and review times are documented estimates; AI runtime is measured.

RISK POSITION

Inherent average	**13.76/25**
Current residual average	**11.94/25**
Target after treatment	**7.24/25**
Evidence available	**25.0% of 68 items**

All values are calculated by `scoring.py` and `evaluate.py` from the master registers (assets, risks, controls, compliance, evidence, audit findings). Control effectiveness counts Implemented controls only; weighted figure treats Partial as 0.5. Compliance uses the A2 weights (Compliant 1, Partial 0.5, Weak 0.25, Non-compliant 0).